

01. Data and Information*

- **Data:**

Raw facts that do not have much meaning on their own. These are values or a set of values. For example, a list of numbers like 45, 67, 89 is just data unless we understand what they represent (marks, ages, temperatures, etc.).

Types of data include:

- Text data (names, addresses)
- Numeric data (marks, salary)
- Image, audio, video, etc.

- **Information:**

When data is processed, organized, or structured in a way that it becomes meaningful and useful, it becomes information.

For example, if we know that 45, 67, 89 are student marks, then it is useful information for performance evaluation.

- **Key Point:**

Information = Processed Data

Data is input, and information is output after processing.

02. What is DBMS and Application

- **DBMS (Database Management System)** is a software system that provides an interface to interact with databases. It allows users to store, modify, retrieve, and manage data efficiently while ensuring security and consistency.
 - **Functions of DBMS:**
 - Creating and organizing the database
 - Inserting, deleting, and updating data
 - Querying the database
 - Managing user access and security
 - Providing backup and recovery
 - **Applications of DBMS:**
 - **Banking Systems** – To manage customer accounts and transactions
 - **Airline and Railway Reservations** – Ticket booking and schedule management
 - **Universities** – Student records, grades, attendance
 - **Hospitals** – Patient records, billing, appointment management
 - **E-commerce** – Product listings, user info, orders
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03. DBMS vs File System*

Feature	File System	DBMS
Data Redundancy	High	Low (due to normalization)
Data Consistency	Difficult to maintain	Easier with constraints and ACID rules
Security	Limited	Advanced (authorization, roles)
Data Integrity	Manual	Enforced by DBMS rules
Backup and Recovery	Manual	Automated by DBMS

- In **file systems**, data is stored in separate files and managed manually.
 - In **DBMS**, everything is centralized, controlled, and efficient.
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04. Define Database Model, Types of Database*

- A **Database Model** is a logical structure that defines how data is stored, organized, and manipulated.

Types of Database Models:

1. **Hierarchical Model:**
 - Data is organized in a tree-like structure (parent-child).
 - Example: File System Directory
 2. **Network Model:**
 - Data is represented by records and relationships are shown using links.
 - More flexible than hierarchical.
 3. **Relational Model** (Most popular):
 - Data is stored in tables (rows and columns).
 - Tables can be related to each other using **keys**.
 - Example: MySQL, PostgreSQL
 4. **Object-Oriented Model:**
 - Combines object-oriented programming with DBMS principles.
 - Stores objects instead of records.
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05. Data Abstraction and Data Independence

- **Data Abstraction:**

DBMS hides complex details from users and shows only the relevant data.

There are **3 levels** of abstraction:

 1. **Physical Level** – How data is physically stored.
 2. **Logical Level** – What data is stored and how they are related.
 3. **View Level** – What part of data users see.
 - **Data Independence:**
 - **Logical Independence:** Changing logical schema without affecting applications.
 - **Physical Independence:** Changing storage without affecting logical schema.
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06. Define Schema and Instances

- **Schema:**
 - Structure or blueprint of the database.
 - Defines tables, columns, data types, constraints.
 - Does **not** change frequently.
- **Instance:**
 - Actual data in the database at a given time.
 - Can **change frequently** as records are inserted, updated, or deleted.

Example:

If you define a `Student` table with (ID, Name, Age), that's the schema.

When you insert records (1, Alice, 20), that's the instance.

07. Compare DML and DDL

Feature	DDL (Data Definition Language)	DML (Data Manipulation Language)
Purpose	Define structure of DB	Manage data inside tables
Common Commands	CREATE, ALTER, DROP	INSERT, UPDATE, DELETE, SELECT
Affect on Data	Structure only	Data only

08. Functional Components of DBMS*

1. **Query Processor:** Interprets and executes database queries (e.g., SQL).
 2. **Storage Manager:** Manages space allocation, file structure, and data storage.
 3. **Transaction Manager:** Ensures transactions follow ACID properties.
 4. **Buffer Manager:** Transfers data between disk and memory efficiently.
 5. **Authorization Manager:** Controls user access and permissions.
 6. **Integrity Manager:** Ensures data integrity rules are followed.
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09. Database Users

1. **Database Administrator (DBA)** – Manages the DBMS, user access, security, and performance.
 2. **Database Designers** – Design the database schema.
 3. **Application Programmers** – Write programs to interact with the database.
 4. **End Users** – Use the database through applications (e.g., students using a portal).
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10. Main Functions of Database Administrator (DBA)*

- Define and modify database schema
 - Provide access controls to users
 - Monitor database performance
 - Take regular backups and manage recovery
 - Ensure security and data integrity
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11. Architecture of DBMS

1. **1-Tier Architecture:**
 - Everything (DBMS + application + user interface) on the same machine.
2. **2-Tier Architecture:**
 - Client interacts directly with the server/database.

3. **3-Tier Architecture:**

- User → Application Server → Database Server
 - Provides better security, scalability, and separation of concerns.
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